Project Report
Of
TYE (Track Your Expenses)

Course: CSE412
Course Title: Software
Section: 02

Submitted To:

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Submitted date: 1 February, 2026

Chapter 1: Project Intro

1.1 Introduction

In an increasingly cashless economy, individuals frequently transact using digital methods such as bKash, Nagad, and bank cards. While convenient, this shift has created a "context gap" where users struggle to track *what* they spent their money on, as standard bank statements often lack categorical details. **TYE (Track Your Expenses)** is an intelligent personal finance application designed to solve this problem by automating expense tracking through Optical Character Recognition (OCR) and AI-driven categorization.

1.2 Overview of the Systems

The TYE system consists of a mobile application (Client) and a backend server (API). The core functionality revolves around a "Universal Parser" engine. Users upload financial documents (PDFs, images), which the system processes to extract transaction data (Date, Amount, Merchant). An intelligent categorization engine then tags these transactions (e.g., "Star Kabab" → "Food"), creating a visual spending dashboard for the user.

1.3 Scope and Objectives

Scope: The project covers the development of a mobile app (Android/iOS) and a backend server. It includes features for uploading transaction proofs (PDF/Image), automated data extraction, manual editing, and visual reporting.

Objectives:

- To reduce manual data entry for expense tracking by 90%.
- To provide accurate categorization of local Bangladeshi transactions (e.g., bKash, local banks).
- To enable users to view real-time spending analytics.

1.4 Stakeholders

- **Primary Users:** Professionals, Students, Business Owners who wants to manage their expenses.
- **Client:** Rafat Bin Islam, Senior Software Engineer, Agoda
- **Development Team:** Team NewToDev, team of four passionate developer (Sifat,Shihab,Adnan and Sabbir)

Chapter 2: Requirements

2.1 Overall Requirement Collection Process

Requirements were gathered through an iterative process involving taking client interview and he suggested to survey the real user of the application(e.g. Job Holders, Students, Bank customers etc.). We have started the survey process. The core focus was on automating the "input" phase of expense tracking to address user fatigue.

2.2 User Stories

ID	Priority	User Story Title	Description
US-01	High	Upload Bank PDF	**User Story 1: Upload Bank Statement (PDF)** As a registered user, I want to upload my monthly bank statement as a PDF file, So that the app can automatically extract all my transaction details without me typing them manually. **Acceptance Criteria:** The user can browse and select a PDF file from their device storage. The system validates that the file is a PDF and under a specific size limit (e.g., 5MB).

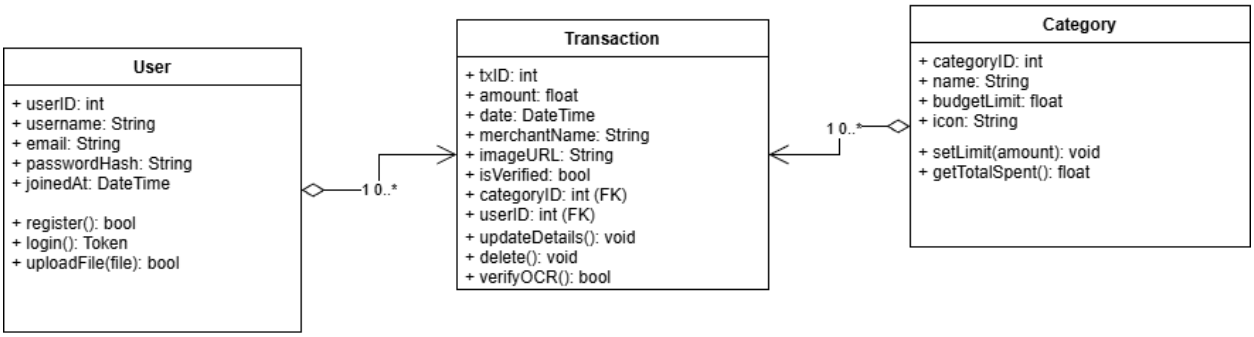
			The system displays a loading indicator while the OCR/Parsing engine processes the file. Upon success, the user sees a "Processing Complete" message and a list of extracted transactions. If the file format is unreadable, an error message guides the user to retry.
US-02	High	Scan Screenshot	**User Story 2: Scan Digital Payment Screenshots** As a user who uses bKash/Nagad, I want to upload screenshots of payment success screens or SMS alerts, So that I can track my digital wallet spending which doesn't appear on my main bank statement. **Acceptance Criteria:** The user can choose to "Take a Photo" or "Select from Gallery." The app allows the user to crop the image to focus on the transaction text. The OCR engine correctly identifies the "Amount," "Date," and "Receiver/Merchant Name" from the image. The system discards irrelevant text (like "Thank you for using bKash") and saves only the transaction data.
US-03	High	Auto-Categorize	**User Story 3: Automatic Categorization** As a user, I want to have my transactions automatically assigned to categories (e.g., Food, Transport), So that I can instantly understand the context of my spending without manual tagging. **Acceptance Criteria:** The system automatically tags known merchants (e.g., "Uber" -> "Transport", "KFC" -> "Food").

			Transactions with unknown merchants are marked as "Uncategorized" by default. The categorization happens immediately after the file/image processing is finished.
US-04	High	Edit Details	**User Story 4: Edit Transaction Details** As a user, I want to manually edit the category or details of an uploaded transaction, So that I can correct any mistakes made by the AI or recategorize an item (e.g., changing a supermarket purchase from "Food" to "Home Supplies"). **Acceptance Criteria**: The user can tap on any specific transaction in the list to open an "Edit" view. The user can change the Category using a dropdown menu. The user can correct the Merchant Name or Amount if the OCR read it wrong. Changes are saved instantly and the main dashboard updates to reflect the new category totals.
US-05	Medium	Visual Dashboard	**User Story 5: Visual Spending Dashboard** As a user, I want to view a pie chart or bar graph of my monthly expenses, So that I can visually compare how much I spent on different categories at a glance. **Acceptance Criteria:** The "Home" tab displays a graphical breakdown of expenses for the current month. The chart uses different colors for different categories (e.g., Red for Food, Blue for Transport).

			Clicking a slice of the pie chart filters the transaction list to show only items in that category. The dashboard shows the "Total Spent" amount clearly at the top.
US-06	Medium	Set Budgets	**User Story 6: Set Monthly Budgets** As a budget-conscious user, I want to set a spending limit for specific categories (e.g., 5000 TK for Food), So that I can track how close I am to overspending for the month. **Acceptance Criteria:** The user can go to a "Budget" section and assign a monetary limit to a category. The system displays a progress bar showing "Spent vs. Limit" (e.g., "3000 / 5000 TK used"). The progress bar turns red if the user exceeds the set limit.
US-07	Low	Search/Filter	**User Story 7: Search and Filter** As a user, I want to search for transactions by keyword or filter by date range, So that I can find a specific payment I made months ago (e.g., verifying if I paid the internet bill). **Acceptance Criteria:** A search bar is available at the top of the transaction list. Entering "Internet" filters the list to show only transactions containing that text. Users can select a "Start Date" and "End Date" to view specific history.
US-08	Low	Export Report	**User Story 8: Export Monthly Report** As a user, I want to download a summary of my

			expenses as a PDF or Excel file, So that I can keep a backup record or share it with my family/accountant. **Acceptance Criteria:** An "Export" button is available on the Dashboard. The user can select the month they want to export. The system generates a clean, readable file listing all transactions and category totals.
US-09	Low	Scan Handwriting	**User Story 9: Scan Handwritten Receipts & Notes** As a user who shops at local markets or writes daily expenses in a diary, I want to take a photo of handwritten receipts or my manual expense list, So that I can digitize these offline expenses and include them in my overall monthly budget without typing them out manually. **Acceptance Criteria:** The user can use the camera to capture an image of handwritten text (e.g., a "memo" from a local shop or a page from a notebook). The system uses advanced OCR to recognize handwritten dates, amounts, and item descriptions. The system presents a "Review Screen" immediately after scanning, asking the user to confirm or correct the text (since handwriting can be messy). Once confirmed, the system successfully assigns categories to these items (e.g., "Vegetables" -> "Groceries") and adds them to the total spending.

2.3 Use Case Diagram:



Chapter 3: Architecture and Detail design

The TYE system follows a three-tier client-server architecture. The presentation layer consists of a mobile or web application that allows users to upload financial documents, view dashboards, and manage budgets. The application layer is implemented using FastAPI, which provides RESTful APIs for handling authentication, file uploads, OCR processing, transaction management, and report generation. The backend integrates with an OCR engine to extract text from PDFs and images and a machine learning module to automatically categorize transactions. The data layer uses PostgreSQL to persist users, transactions, budgets, and category information. This layered architecture ensures scalability, maintainability, and separation of concerns.

Appendix:

1. **Project Information Drive LInk**
2. **TYE(Track Your Expenses) Github**
3. **Initial Project Proposal**